

STA 2002

Lecture 13

Order statistics

Order statistics

$X_1, X_2, \cdots X_n$ are independent random variables with the common cdf F and density f .

Denote the ***ordered*** sample values by $X_{(1)} < X_{(2)} \cdots < X_{(n)}$. These values are called **order statistics.**

THEOREM **A**

The density of $X_{(k)}$, the kth-order statistic, is

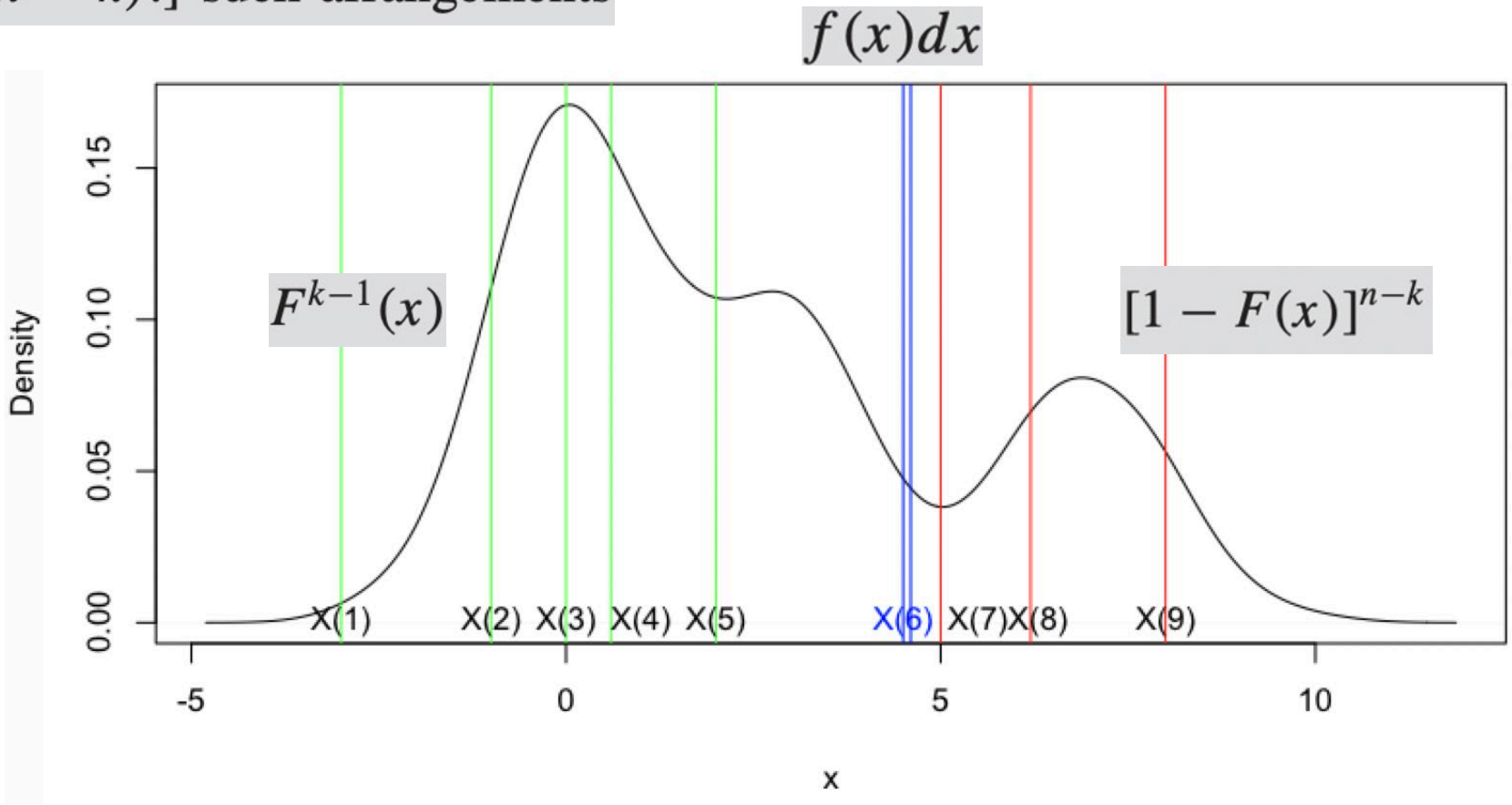
$$f_k(x) = \frac{n!}{(k-1)!(n-k)!} f(x) F^{k-1}(x)[1 - F(x)]^{n-k}$$

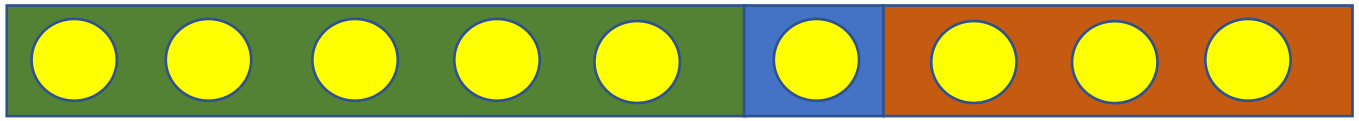
The event $x \leq X_{(k)} \leq x + dx$ occurs if $k - 1$ observations are less than x, one observation is in the interval $[x, x + dx]$, and $n - k$ observations are greater than $x + dx$. The probability of any particular arrangement of this type is $f(x) F^{k-1}(x)[1 - F(x)]^{n-k} dx$, and, by the multinomial theorem, there are $n!/[(k-1)!1!(n-k)!]$ such arrangements, which completes the argument. ■

Order statistics

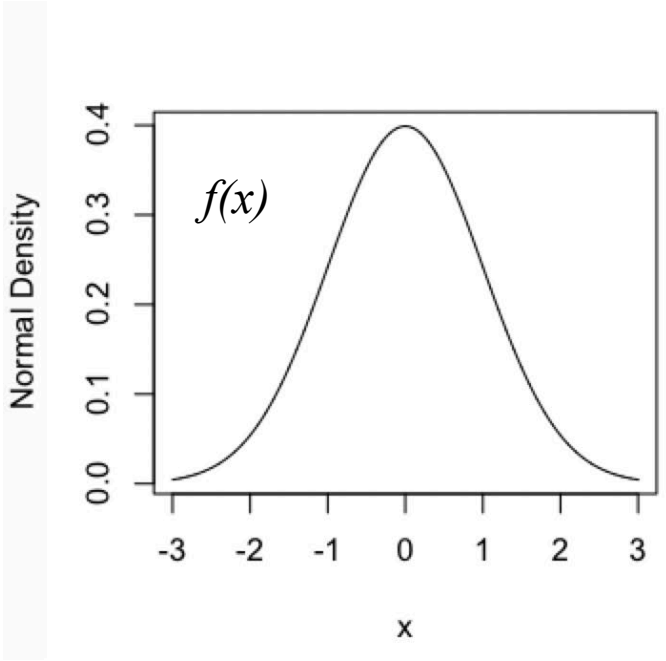
$$f_k(x) = \frac{n!}{(k-1)!(n-k)!} f(x) F^{k-1}(x)[1-F(x)]^{n-k}$$

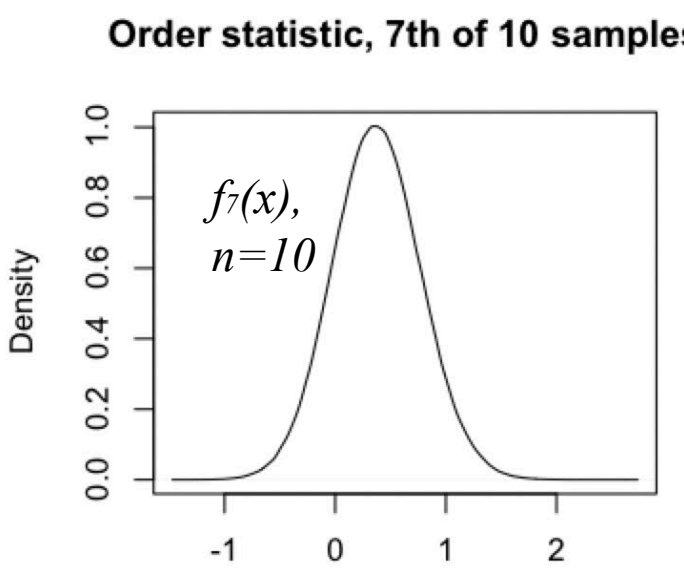
$n!/[(k-1)!1!(n-k)!]$ such arrangements

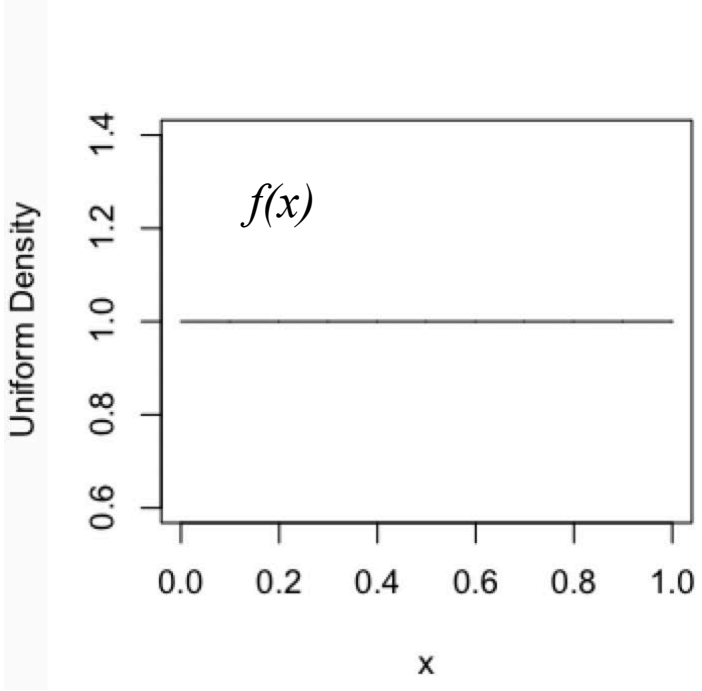


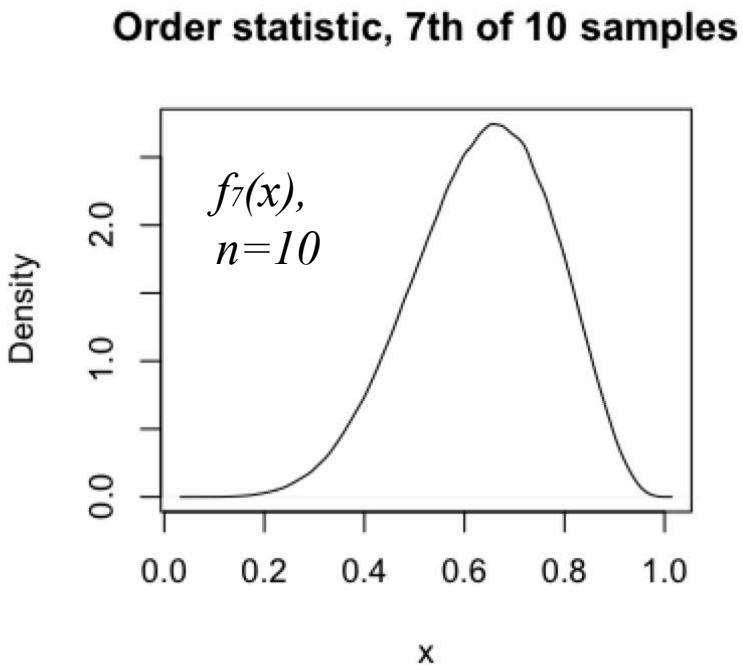


Order statistics

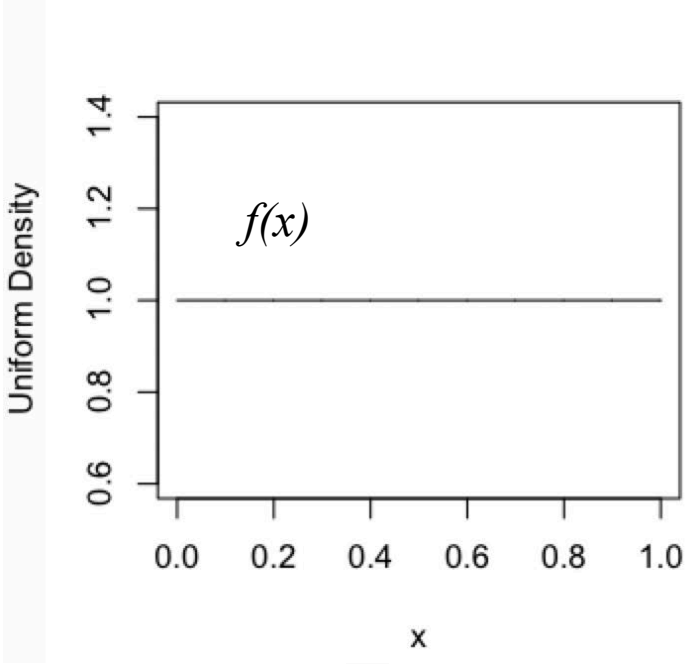


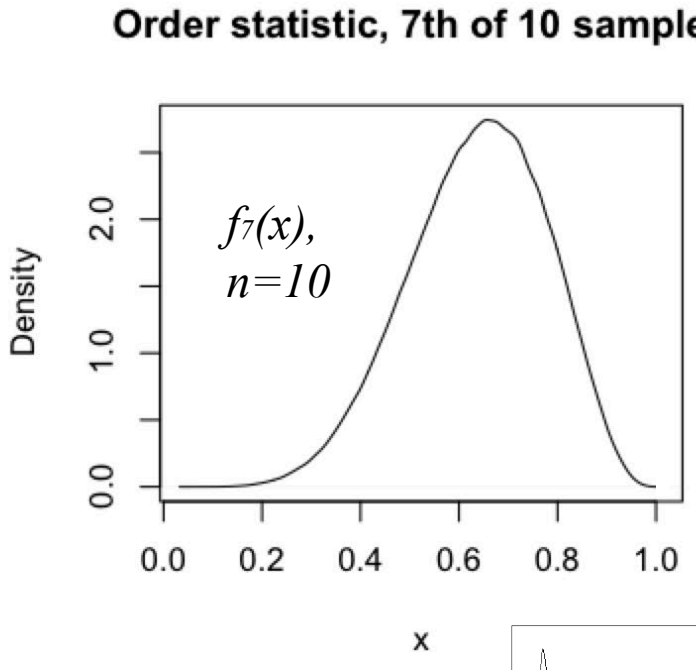


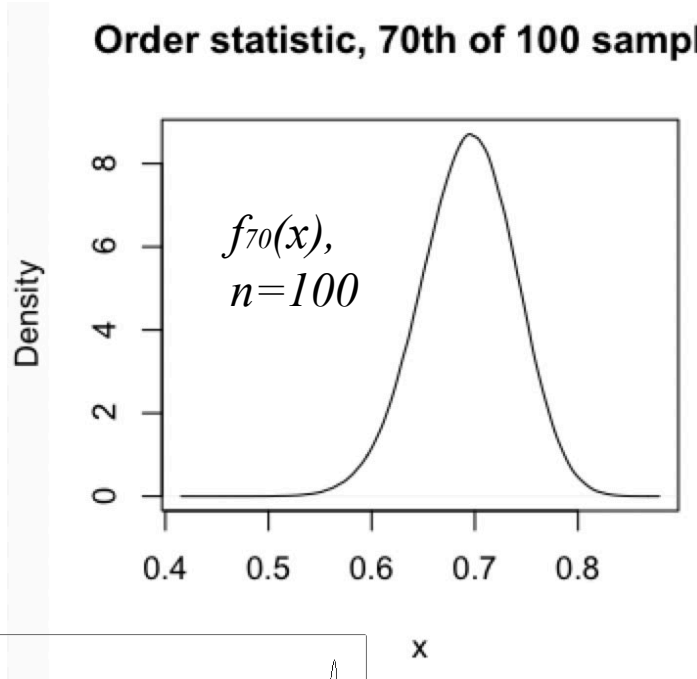


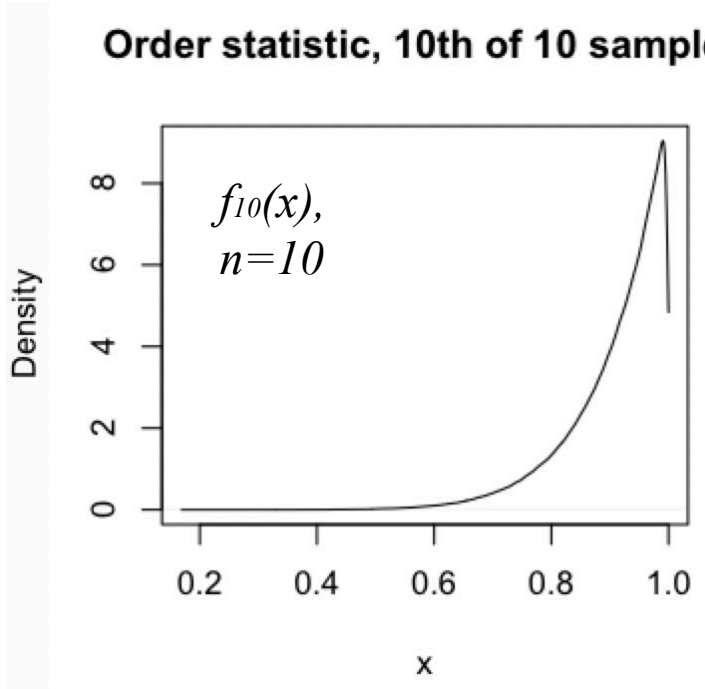


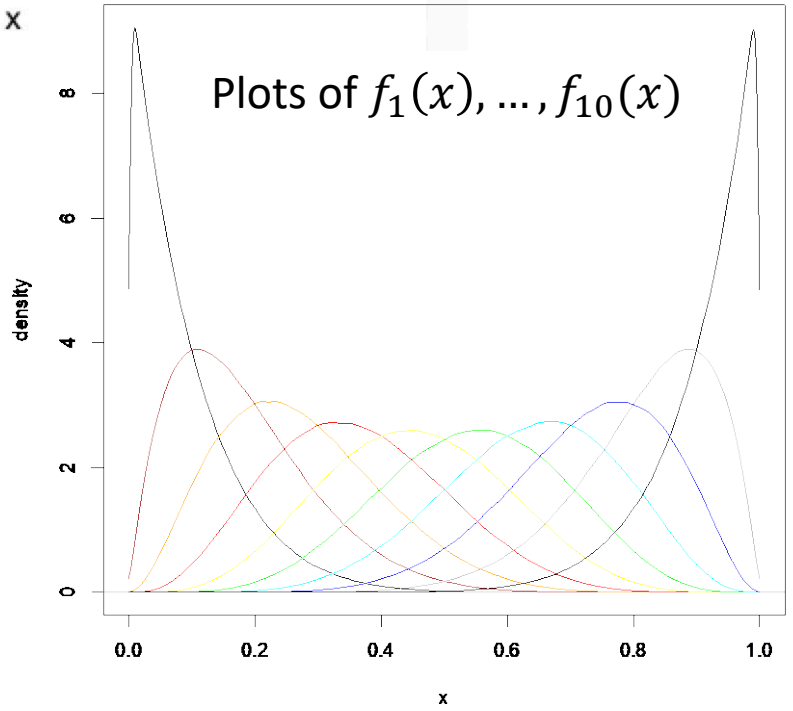
Order statistics











Order statistics

For the case where the X_i are uniform on [0, 1], the density of the kth-order statistic reduces to

$$f_k(x) = \frac{n!}{(k-1)!(n-k)!} f(x) F^{k-1}(x)[1 - F(x)]^{n-k}$$

$f(x)=1$
$F(x)=x$

$$\frac{n!}{(k-1)!(n-k)!} x^{k-1}(1-x)^{n-k}, \qquad 0 \le x \le 1$$

This is the density of beta-distribution.

$$\alpha = k$$
$$\beta = n - k + 1$$

PDF

$$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{\mathrm{B}(\alpha,\beta)}$$

where $\mathrm{B}(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ and Γ is the Gamma function.

$\Gamma(n) = (n-1)!$ when n is positive integer

Order statistics

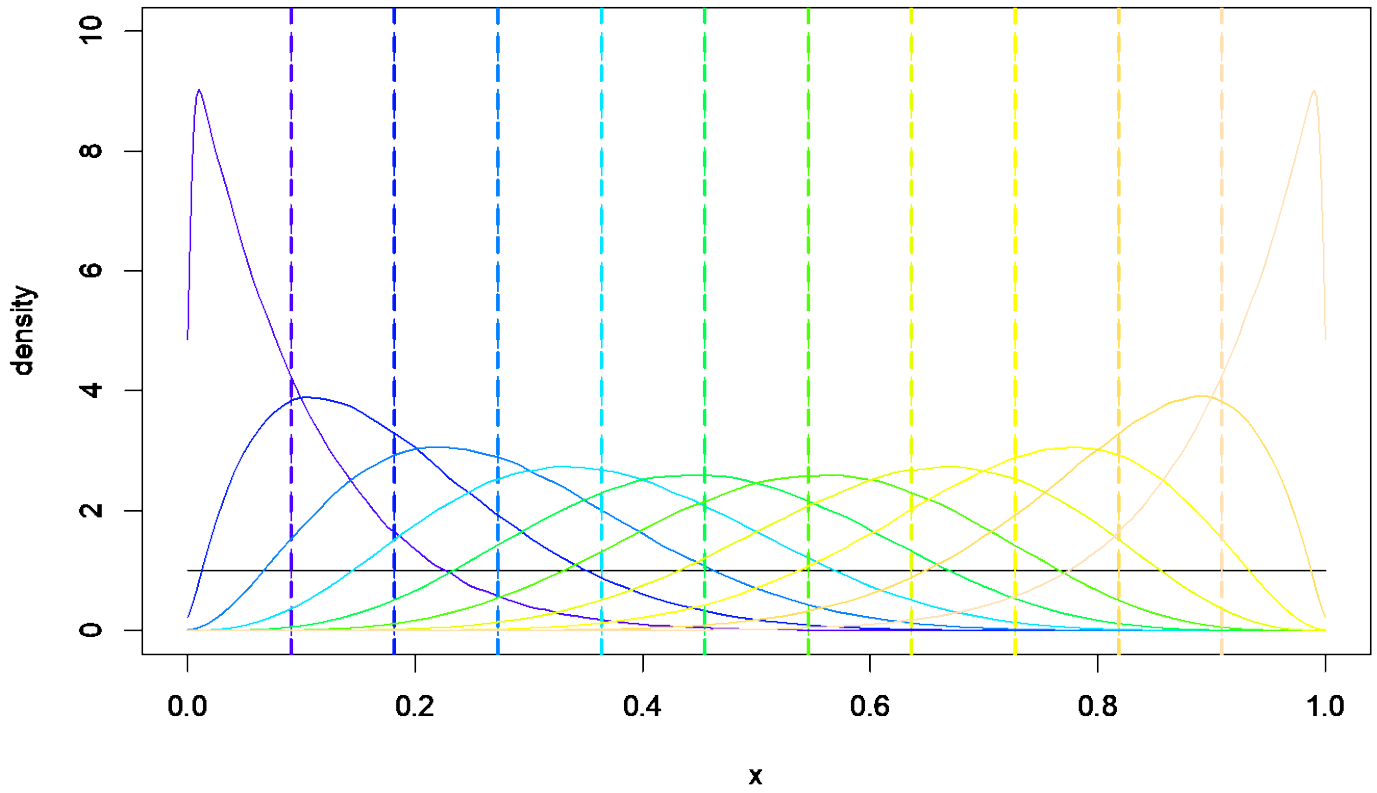
PDF	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{\mathrm{B}(\alpha,\beta)}$ where $\mathrm{B}(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ and Γ is the Gamma function.

$$\alpha = k$$

$$\beta = n - k + 1$$

$$\mathrm{E}[X_{(k)}] = \frac{\alpha}{\alpha+\beta} = \frac{k}{n+1}$$

The vertical lines are 1/11,…,10/11 for n=10.



Order statistics

The pth **quantile** of the distribution F is defined to be that value x_p such that $F(x_p) = p$, or $P(X \leq x_p) = p$.

$$F(x) = p$$

$$x_p = F^{-1}(p)$$

p = 1/2, corresponds to the **median** of F;

p =1/4 and p =3/4, correspond to the lower and upper **quartiles** of F.

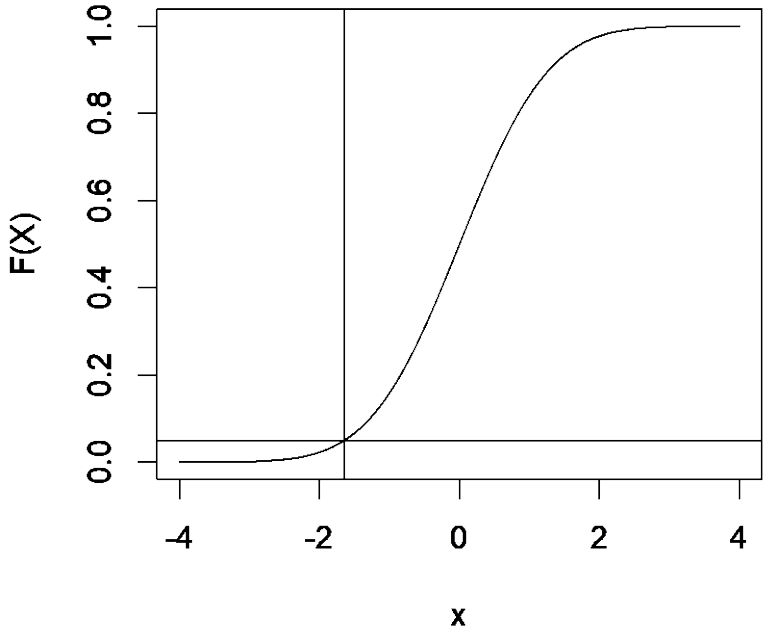
Ex:

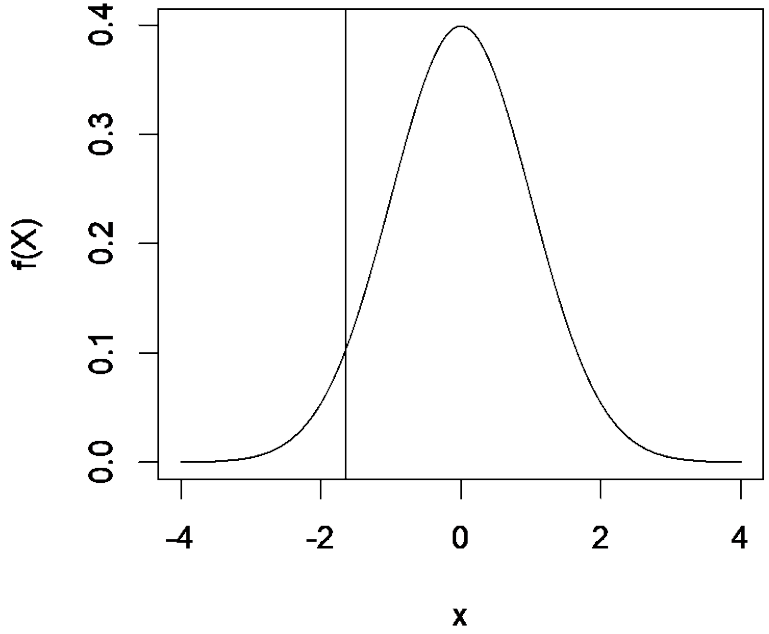
F is the CDF of N(0,1)

$F(-1.645) = 0.05$

$-1.645 = F^{-1}(0.05)$

$$\int_{-\infty}^{-1.645} f(x)dx = 0.05$$





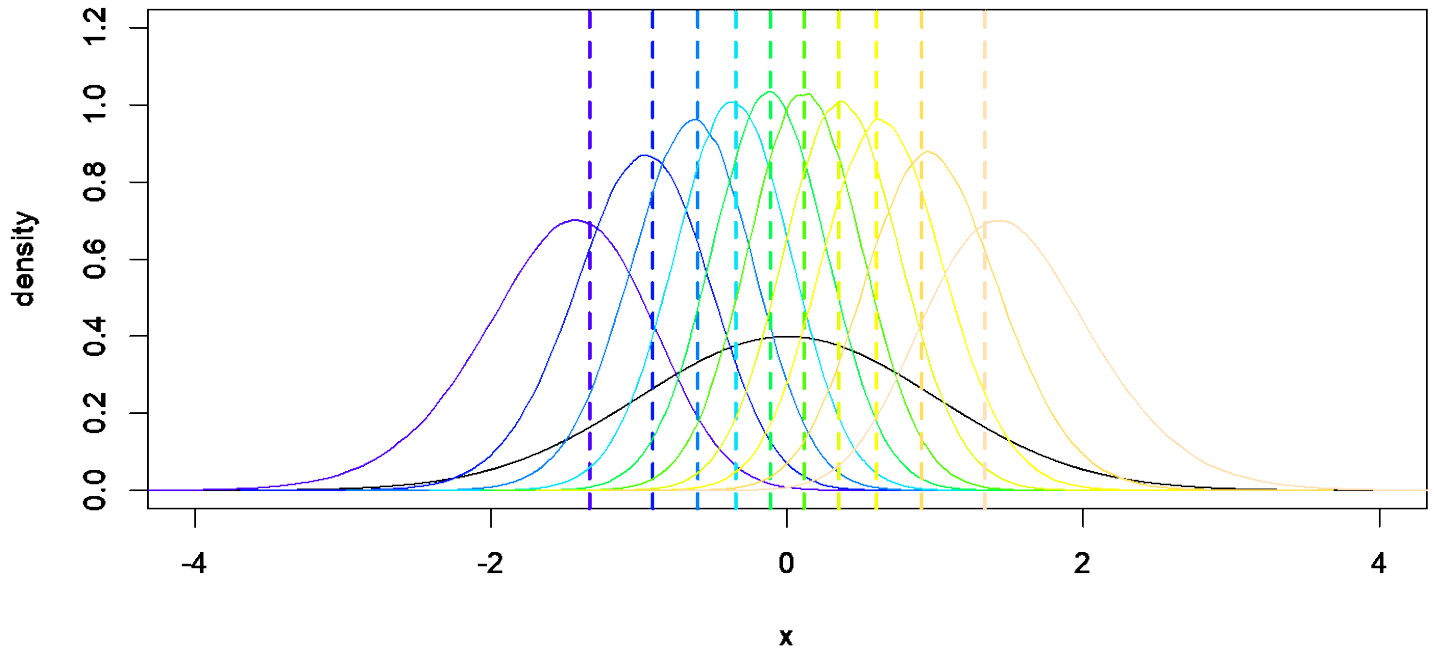
Order statistics

For a uniform distribution, the expectation of the kth order statistic from n samples is equal to $\frac{k}{n+1}$, which is the $\frac{k}{n+1}$ -*th* quantile of the distribution because $F(\frac{k}{n+1}) = \frac{k}{n+1}$, i.e., $F(EX_{(k)}) = \frac{k}{n+1}$.

Is this true for other distributions? Not necessarily.

Ex:

N(0,1) distribution:
Curves: distribution of sample order statistics (n=10)
Vertical bars: $\frac{k}{10+1}$ -*th* quantile of the distribution, which my be different from the expectation of the kth order statistic .



Order statistics

PROPOSITION **C**

Let $Z = F(X)$; then Z has a uniform distribution on [0, 1].

Proof

$$P(Z \leq z) = P(F(X) \leq z) = P(X \leq F^{-1}(z)) = F(F^{-1}(z)) = z$$

This is the uniform cdf. ■

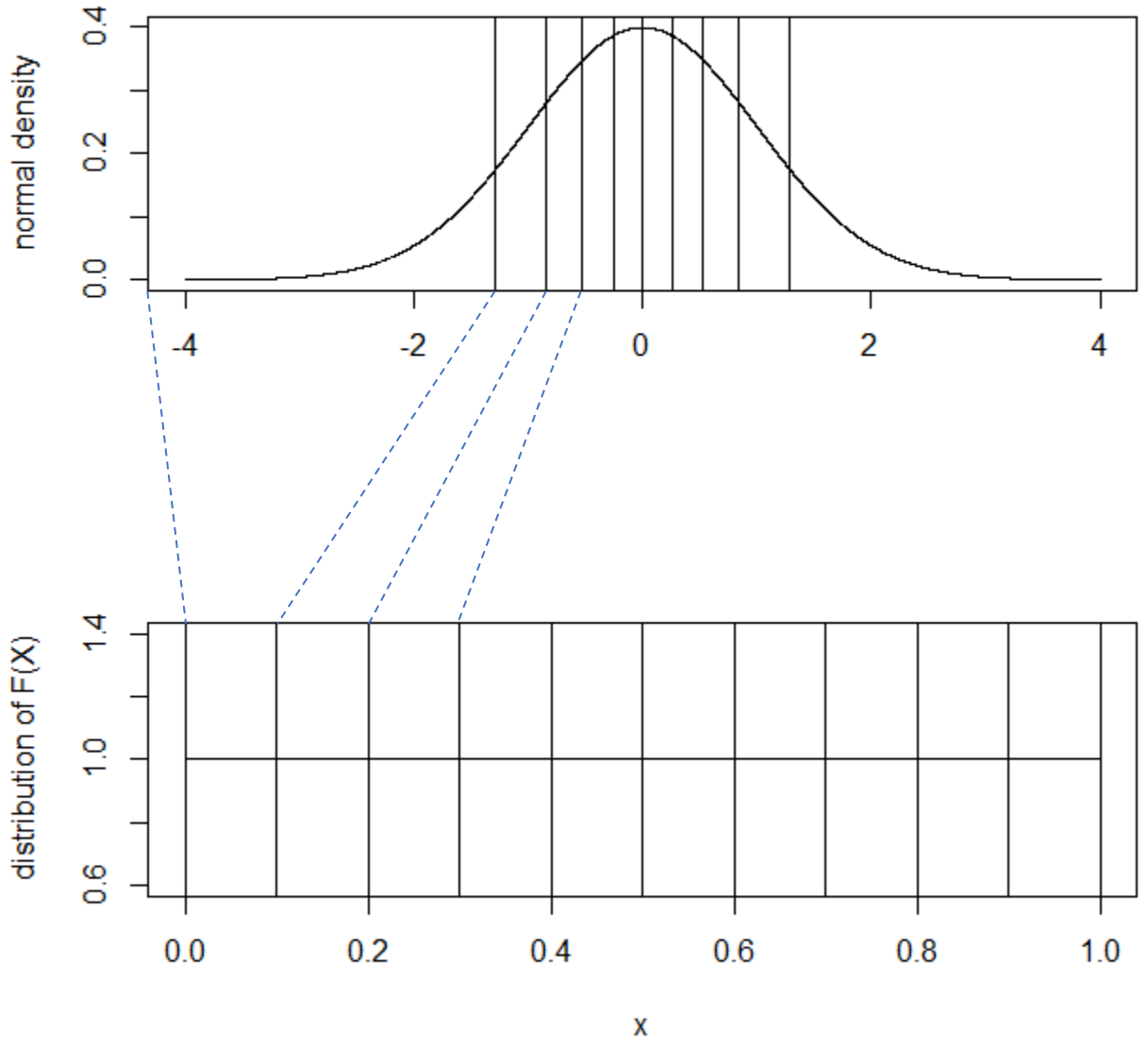
PROPOSITION **D**

Let U be uniform on [0, 1], and let $X = F^{-1}(U)$. Then the cdf of X is F.

Proof

$$P(X \leq x) = P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x)$$ ■

$F^{-1}(0.1), F^{-1}(0.2), \ldots\ldots, F^{-1}(0.9)$



Order statistics

Reminder:

$X_1, X_2, \cdots X_n$ are independent random variables with the common cdf F and density f

Then,

$F(X_1), F(X_2) \cdots F(X_n)$ follow U[0, 1] distribution

In addition,

$F(X_{(1)}), F(X_{(2)}) \cdots F(X_{(n)})$ maintains the same increasing order as $X_{(1)}, X_{(2)} \cdots X_{(n)}$

Let U=F(X)

For U, the expectation of the kth order statistic from n samples is equal to the $\frac{k}{n+1}$ - quantile of the distribution.

$$\rightarrow \quad E\left(U_{(k)}\right) = \frac{k}{n+1} \quad \rightarrow \quad E\left(F(X_{(k)})\right) = \frac{k}{n+1}$$

Quantile-Quantile plot

Plotting sample quantiles against quantiles of a theoretical distribution can help us judge if the sample is likely to come from such a distribution.

If *X*~*F*, we know $E\left(F(X_{(k)})\right) = \frac{k}{n+1}$

This means we should plot $F(X_{(k)})$ vs. $\frac{k}{n+1}$

So, if *X*~*F* is indeed true, a plot of $F\left(X_{(k)}\right)$ vs $\frac{k}{n+1}$ should be close to a 45 degree straight line.

Alternatively and usually, we plot $X_{(k)}$ vs. $F^{-1}\left(\frac{k}{n+1}\right)$

k^{th} ordered data point

$\frac{k}{n+1}$ quantile of the distribution

Quantile-Quantile plot

Problem: the expectation of $X_{(k)}$ is not exactly at the quantile $F^{-1}\left(\frac{k}{n+1}\right)$.

What we have is $E\left(F(X_{(k)})\right) = \frac{k}{n+1}$, but this does not imply $E\left(X_{(k)}\right) = F^{-1}\left(\frac{k}{n+1}\right)$.

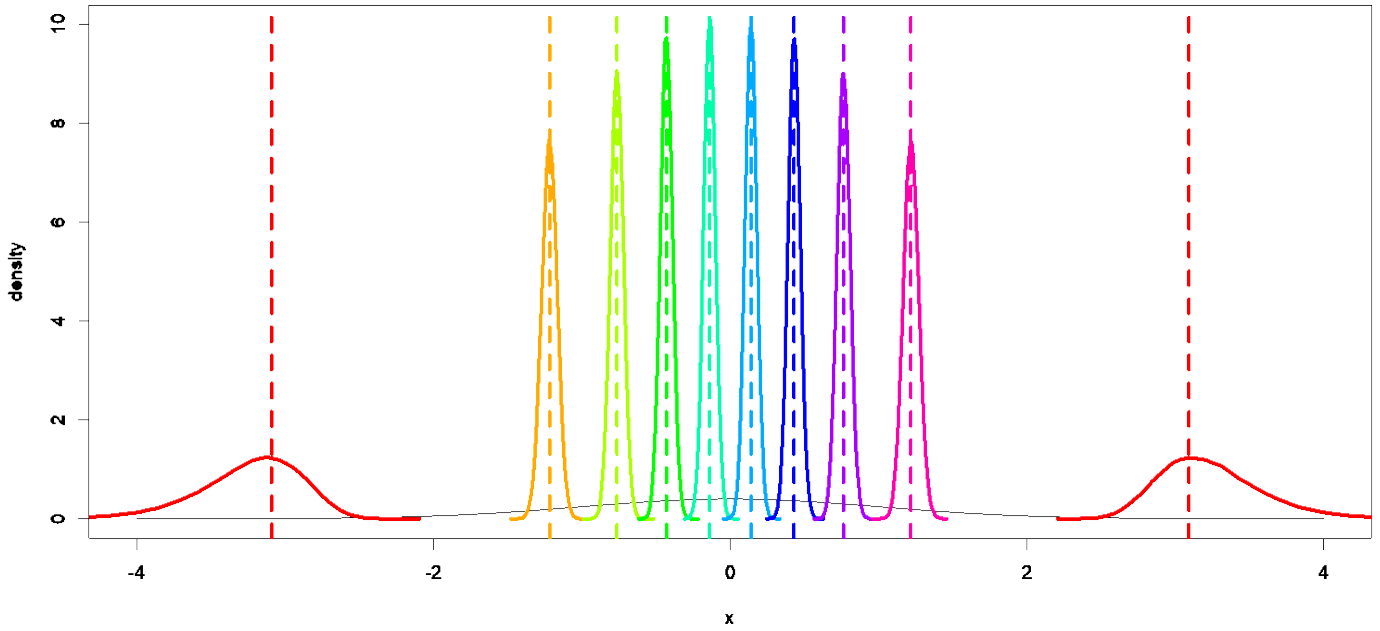
However, they should be pretty close if n is large.

1000 samples;

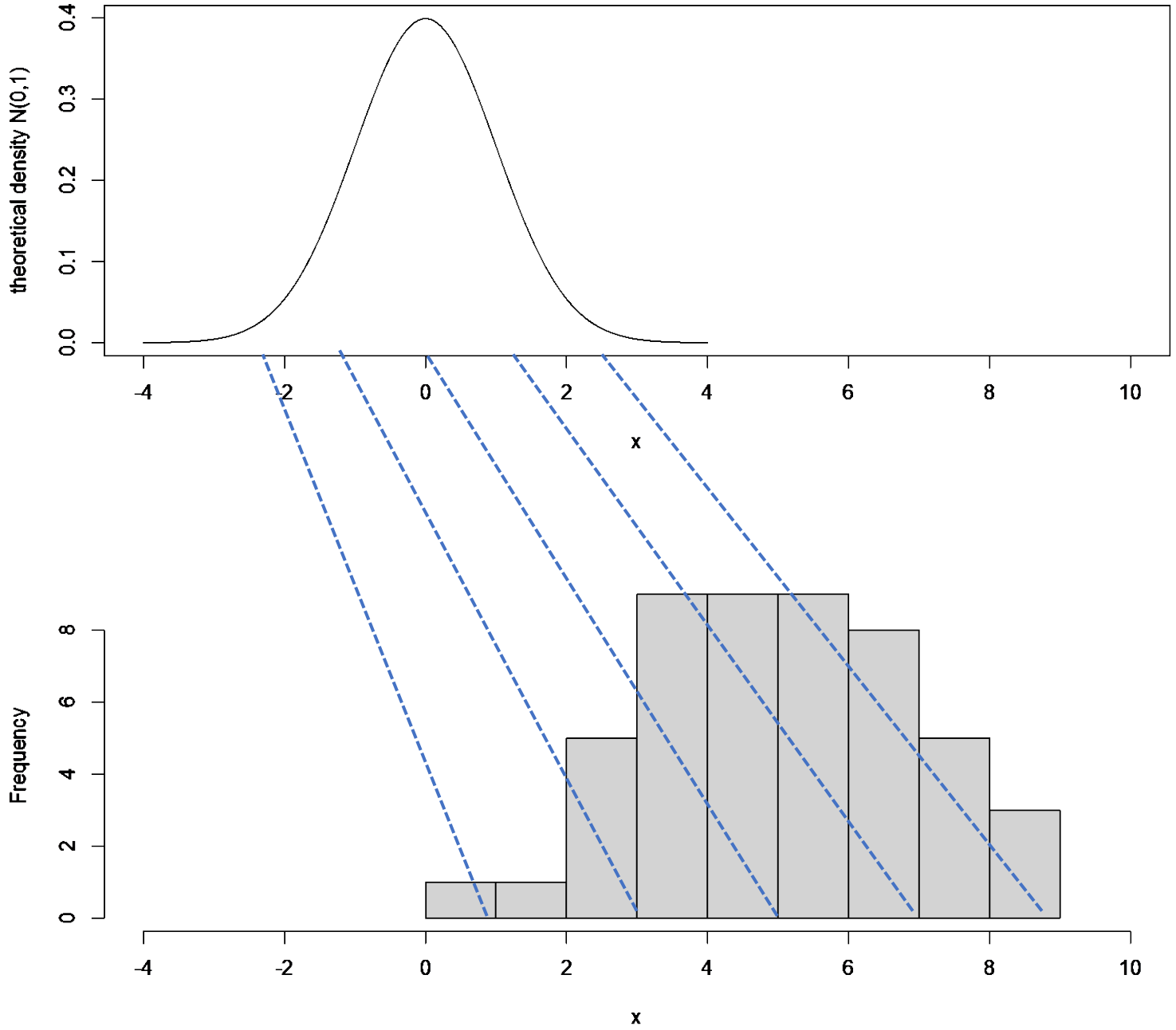
k= 1 112 223 334 445 556 667 778 889 1000

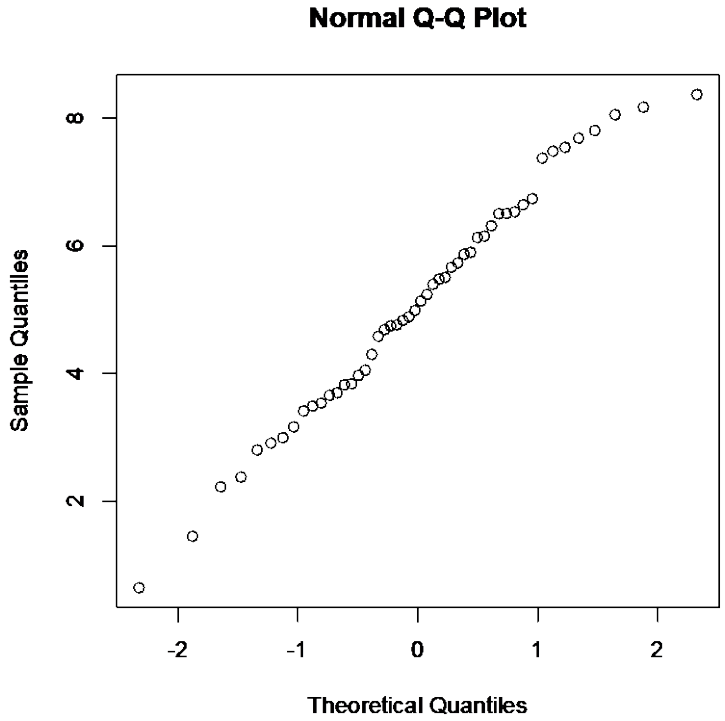
Average $X_{(1)}$: -3.24
$F^{-1}(1/1001)$: -3.09

Average($F(X_{(1)})$) ≈ 1/1001



EX The distributions can be shifted and stretched. As long as the shapes match, the Q-Q plot is close to a straight line.





EX

This is not a rigorous test. But when Q-Q plot against the theoretical values is close to a straight line, the data is likely to follow such a distribution.

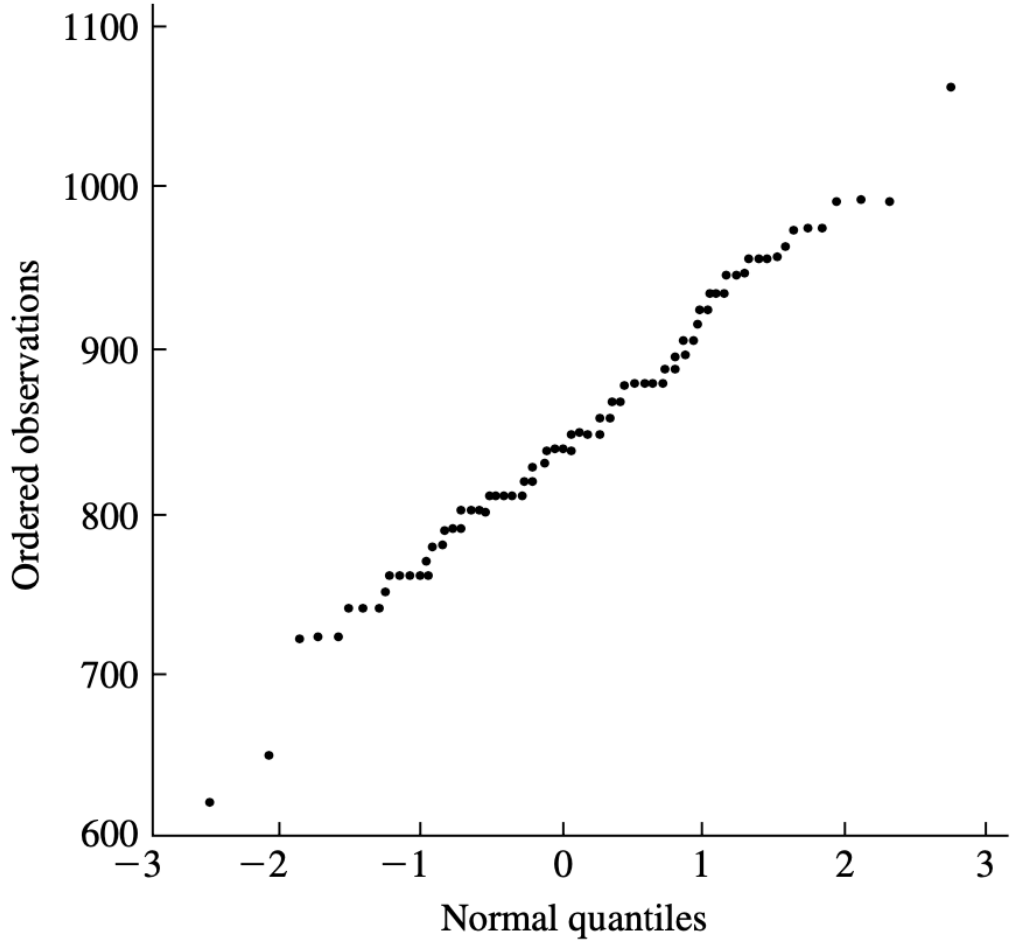
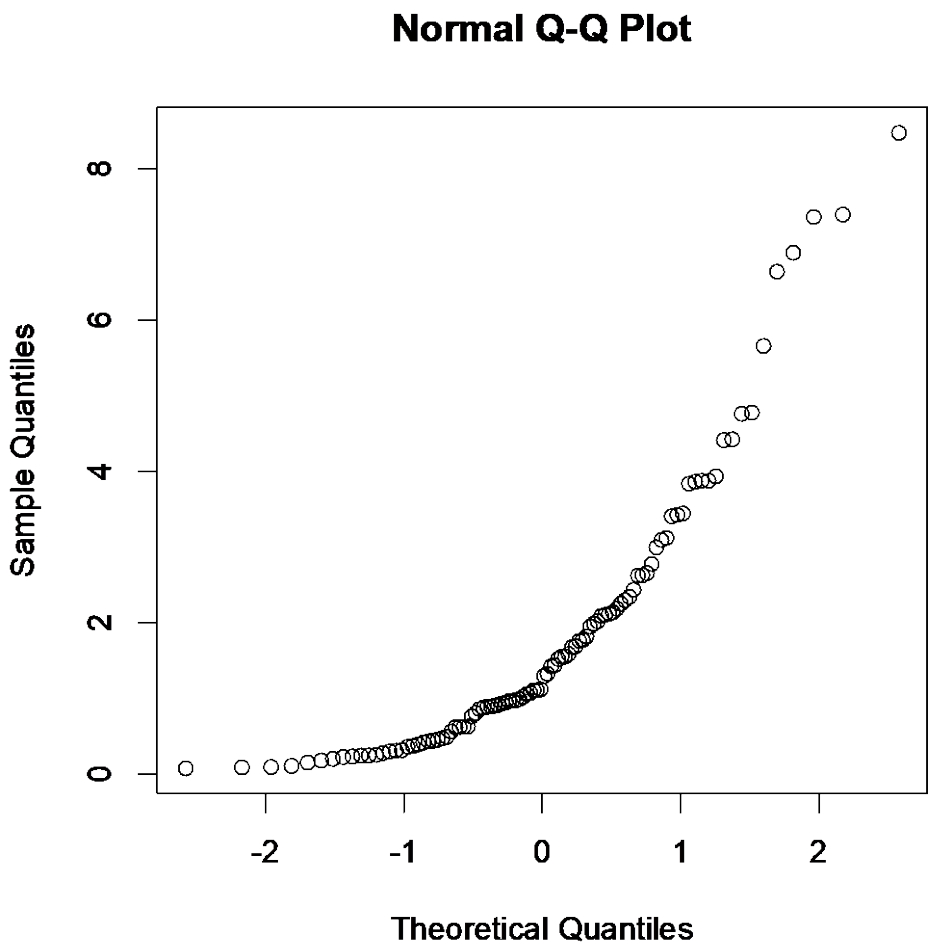
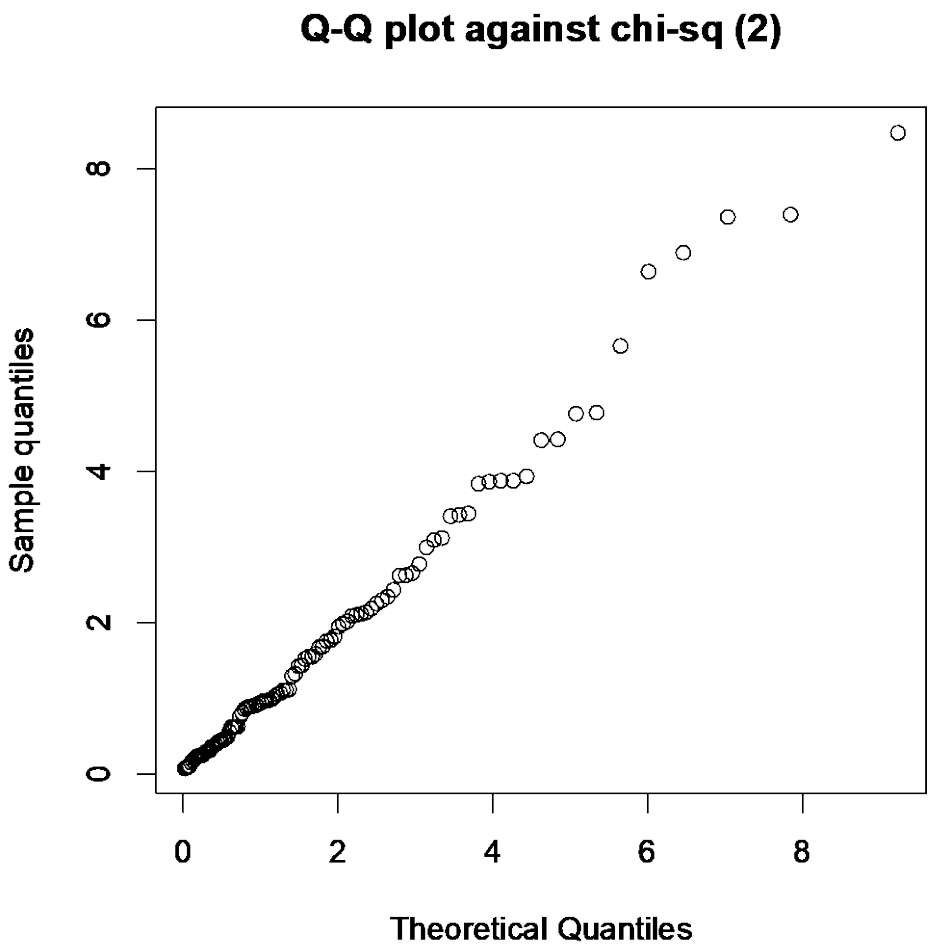


FIGURE **9.6** Normal probability plot of Michelson's data.

EX

some data that follows χ_2^2





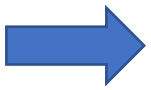
Quantile-Quantile (QQ) plot



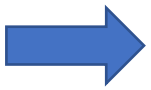
In a Q-Q plot, the quantiles of one distribution are plotted against those of another.

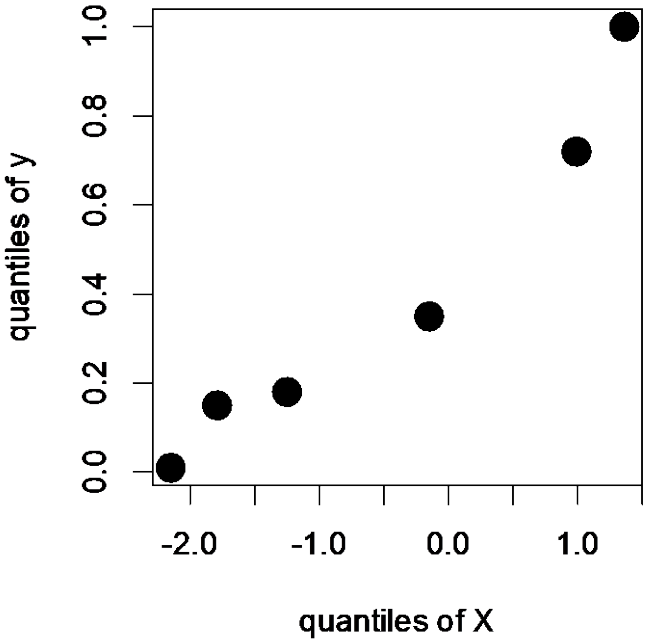
Usually at least one of the distributions is an **empirical distribution** (usually a collection of i.i.d. samples)

x_1	0.99	y_1	0.18
x_2	-1.25	y_2	0.01
x_3	-2.15	y_3	0.35
x_4	-0.15	y_4	0.72
x_5	1.36	y_5	1.00
x_6	-1.79	y_6	0.15



$x_{(1)}$	-2.15	$y_{(1)}$	0.01
$x_{(2)}$	-1.79	$y_{(2)}$	0.15
$x_{(3)}$	-1.25	$y_{(3)}$	0.18
$x_{(4)}$	-0.15	$y_{(4)}$	0.35
$x_{(5)}$	0.99	$y_{(5)}$	0.72
$x_{(6)}$	1.36	$y_{(6)}$	1.00

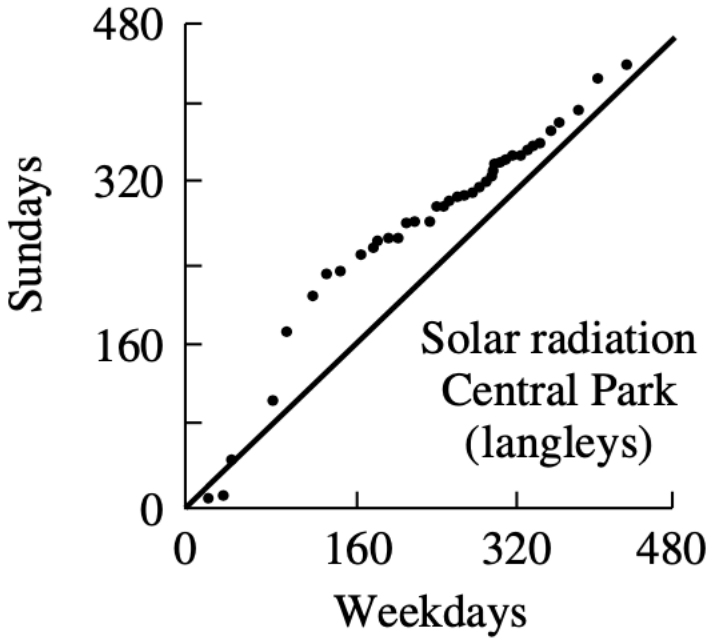




Because $X_{(1)}, \ldots, X_{(n)}$ divide the data range into $(n+1)$ pieces, we treat $X_{(k)}$ as the $\left(\frac{k}{n+1}\right)$th empirical quantile.

Q-Q plot
two samples

Q-Q plots in a study of air pollution. Distributions of solar radiation on Sunday against the quantiles for weekdays.



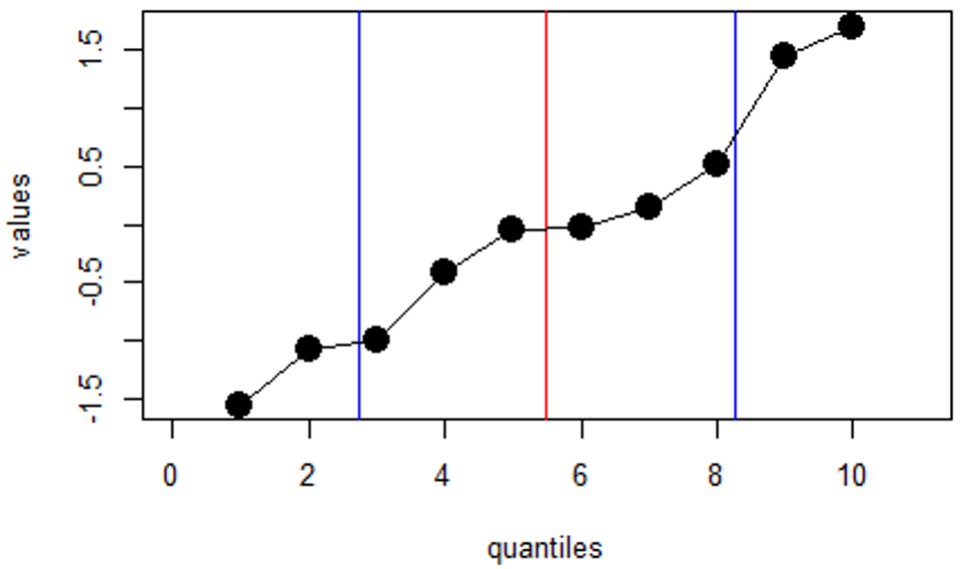
"The very high and very low quantiles of solar radiation are about the same on Sundays and weekdays (presumably corresponding to very clear days and days with heavy cloud cover), but for intermediate quantiles, the Sunday quantiles are larger."

Order statistics

Finding median and quartiles.

For example, how do we find the third quartile, i.e., the $\left(\frac{3}{4}\right)$th quantile, with 10 data points?

Recall $X_{(k)}$ is the $\left(\frac{k}{n+1}\right)$th empirical quantile. Thus, $X_{(8)}$ and $X_{(9)}$ are the $\left(\frac{8}{11}\right)$th and $\left(\frac{9}{11}\right)$th empirical quantiles. Let y denote the third quartile, and let y_1 and y_2 denote $X_{(8)}$ and $X_{(9)}$. Since ¾ is between 8/11 and 9/11, we calculate y as following.



$$\frac{y_2 - y_1}{\frac{9}{11} - \frac{8}{11}} = \frac{y - y_1}{\frac{3}{4} - \frac{8}{11}}$$

$$\frac{3}{4}y_2 - \frac{8}{11}y_2 - \frac{3}{4}y_1 + \frac{8}{11}y_1 = \frac{1}{11}y - \frac{1}{11}y_1$$

$$y_2 - y_1 = 4y - 4y_1$$

$$y = \frac{1}{4}y_2 + \frac{3}{4}y_1$$